

Julien Siems



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EDUCATION

University of Freiburg

Masters in Computer Science; GPA: 4.0 Graduated with Distinction

Courses: Numerical Optimization, Optimal Control, Automated Machine Learning, Game Theory

Freiburg, Germany

Oct 2018 – Mar 2021

University College London

Erasmus Exchange Student

Courses: Probabilistic Graphical Models, Image Processing, Information Retrieval, Computer Vision

London, United Kingdom

Sep 2016 – Jun 2017

TU Dresden

Bachelors in Computer Science; GPA: 3.5

Dresden, Germany

Oct 2014 – Dec 2017

École Polytechnique Fédérale de Lausanne (EPFL)

First year courses in Mechanical Engineering

Lausanne, Switzerland

Sept 2013 – Jul 2014

EXPERIENCE

Microsoft Research

Research Intern

Cambridge, UK

Sept 2025 - current

University of Freiburg

PhD Student – Machine Learning Group supervised by Prof. Frank Hutter

Freiburg, Germany

Nov 2023 - current

- Focus: I am primarily focusing on the in-context learning and state-tracking abilities of linear RNNs such as Mamba and DeltaNet
- Organizer of the In-Context Learning Workshop at ICML 2024

Merantix Momentum

Machine Learning Researcher

Berlin, Germany

Jun 2022 – Aug 2023

- NeurIPS 2023 conference paper on mitigating interpretability issues caused by concavity in generalized additive models: This project focused on addressing concavity, the non-linear equivalent of multicollinearity, which arises in the application of Generalized Additive Models, more specifically Neural Additive Models and Neural Prophet.
- Interpretable Additive Time-Series Forecasting: We modified Neural Prophet, an interpretable additive time-series forecasting method, to allow for complex relationships between external time-series. This allowed us to win the Solar Energy Forecasting Challenge hosted by Hessian AI in February 2023.
- Interpretable Reinforcement Learning: In this project, I worked on interpretable reinforcement learning in the context of inventory management. The aim was to design an agent which is more flexible than classical static optimization methods, yet is fully interpretable.

TU Munich

Research Assistant – Machine Learning and Inverse Problems Group

Munich, Germany

Nov 2021 - Apr 2022

- Adversarial training for inverse problems: My project focused on optimization of inverse problems when faced with provably maximally adversarial attacks. My contributions were the mathematical and empirical analysis of the problem for a linear model.

University of Zurich

Research Assistant – Computation and Economics Research Group

Zurich, Switzerland (Remote from Berlin)

Apr 2021 – Sept 2021

- Iterative Combinatorial Auctions: As part of my stay at the computational economics group, I co-authored several publications on Machine Learning powered Iterative Combinatorial Auctions. These are auctions which are heavily used in mobile frequency auctions such as 5G or 6G.

Amazon Web Services

Applied Science Intern

Berlin, Germany

Sep 2020 – Mar 2021

- Dynamic Neural Network Pruning: My internship in the AWS Sagemaker Team focused on AutoML and Dynamic Network Pruning. Our proposed neural network weight pruning method can be used to make neural networks more efficient to evaluate and save energy costs.

PUBLICATIONS (* DENOTES EQUAL CONTRIBUTION)

- Vladyslav Moroshav*, Julien Siems*, Arber Zela, Timur Carstensen*, Frank Hutter. "TempoPFN: Synthetic Pre-training of Linear RNNs for Zero-shot Time Series Forecasting". *AI in Tabular Data Workshop at EurIPS 2025*.
- Julien Siems*, Timur Carstensen*, Arber Zela, Frank Hutter, Massimiliano Pontil, Riccardo Grazi*. "DeltaProduct: Improving State-Tracking in Linear RNNs via Householder Products". *NeurIPS 2025*.
- Riccardo Grazi*, Julien Siems*, Jörg Franke, Arber Zela, Frank Hutter, Massimiliano Pontil. "Unlocking State-Tracking in Linear RNNs through Negative Eigenvalues". *Modern Mathematics of Machine Learning Workshop at NeurIPS 2024*.
- Sathya Kamesh Bhethanabhotla*, Omar Swelam*, Julien Siems, David Salinas, Frank Hutter. "Mamba4Cast: Efficient Zero-Shot Time Series Forecasting with State Space Models". *Workshop on Time Series in the Age of Large Models at NeurIPS 2024*.
- Andreas Müller*, Julien Siems*, Harsha Nori, David Salinas, Arber Zela, Rich Caruana, Frank Hutter. "GAMformer: In-Context Learning for Generalized Additive Models". *Interpretable AI Workshop & Tabular Representation Learning Workshop at NeurIPS 2024*.
- Riccardo Grazi*, Julien Siems*, Simon Schrodi, Thomas Brox, Frank Hutter. "Is Mamba Capable of In-Context Learning?". *ME-FOMO Workshop at ICLR 2024 & AutoML Conference 2024*.
- Julien Siems*, Konstantin Ditschuneit*, Winfried Ripken*, Alma Lindborg*, Maximilian Schambach, Johannes S. Otterbach, Martin Genzel. "Curve Your Enthusiasm: Concurvity Regularization in Differentiable Generalized Additive Models". *Advances in Neural Information Processing Systems 37 (NeurIPS) 2023*.
- Julien Siems, Maximilian Schambach, Sebastian Schulze and Johannes Otterbach. "Interpretable Reinforcement Learning via Neural Additive Models for Inventory Management." *AI4ABM Workshop at International Conference on Learning Representations (ICLR) 2023*.
- Jakob Weissteiner*, Jakob Heiss*, Julien Siems* and Sven Seuken. "Bayesian Optimization-based Combinatorial Assignment." *AAAI 2023*.
- Jakob Weissteiner*, Jakob Heiss*, Julien Siems* and Sven Seuken. "Monotone-value neural networks: Exploiting preference monotonicity in combinatorial assignment." *International Joint Conference on Artificial Intelligence (IJCAI) 2022*.
- Arber Zela*, Julien Siems*, Lucas Zimmer*, Jovita Lukasik, Margret Keuper and Frank Hutter. "Surrogate NAS Benchmarks: Going Beyond the Limited Search Spaces of Tabular NAS Benchmarks." *International Conference on Learning Representations (ICLR) 2022*.
- Julien Siems, Aaron Klein, Cedric Archambeau and Maren Mahsereci. "Dynamic Pruning of a Neural Network via Gradient Signal-to-Noise Ratio." *8th ICML Workshop on Automated Machine Learning (AutoML) at International Conference On Machine Learning (ICML) 2021*.
- Arber Zela*, Julien Siems* and Frank Hutter. "NAS-Bench-1Shot1: Benchmarking and Dissecting One-shot Neural Architecture Search." *International Conference on Learning Representations (ICLR) 2020*.
- Roland Zimmermann* and Julien Siems*. "Faster training of Mask R-CNN by focusing on instance boundaries." *Computer Vision and Image Understanding 188 (2019): 102795*.

PATENTS

- Arber Zela, Frank Hutter, Julien Siems, Lucas Zimmer, Method, device and computer program for predicting a suitable configuration of a machine learning system for a training data set, US Patent No. 16950570.

- Arber, Zela, Frank Hutter, Julien Siems, Lucas Zimmer, Verfahren, Vorrichtung und Computerprogramm zum Vorhersagen einer geeigneten Konfiguration eines maschinellen Lernsystems für einen Trainingsdatensatz, German Patent No. DE: 10 2020 202 392.1.
- Julien Siems, Arber Zela, Frank Hutter, Verfahren, Vorrichtung und Computerprogramm zum Erstellen eines maschinellen Lernsystems, German Patent No. DE: 10 2019 214 500.0.

SKILLS SUMMARY

- **Languages:** Python, C++
- **Tools:** PyTorch, Jax, Scipy, Numpy, Docker, git, Matlab